

```
//jayneil.dalal@gmail.com, 18-6-12
#include "opencv2/imgproc/imgproc.hpp"
#include "opencv2/highgui/highgui.hpp"
#include <iostream>
using namespace cv;
using namespace std;
int main(int argc, char** argv) {
    VideoCapture capture(CV_CAP_OPENNI);
    for(;;) {
        //variable declaration
        Mat bgrImage;
        capture.grab();
        capture.retrieve_(<_bgrImage, CV_CAP_
OPENNI_BGR_IMAGE );
        //Display rgb image
        imshow( "rgb", bgrImage);
        if( waitKey( 30 ) >= 0 )
            break;
    }
    return 0;
}
```

Example 2: Display depth data from Kinect in OpenCV (see Figure 2):

```
//Program to display the depth map in
OpenCV 2.4.2 using Kinect
//jayneil.dalal@gmail.com, 18-6-12
#include "opencv2/imgproc/imgproc.hpp"
#include "opencv2/highgui/highgui.hpp"
#include "opencv2/opencv.hpp"
#include <iostream>
using namespace cv;
using namespace std;
int main(int argc, char** argv) {
    VideoCapture capture( CV_CAP_OPENNI );
    for(;;) {
        //variable declaration
        Mat depthMap;
        Mat show;
        const float scaleFactor = 0.05f;
        capture >> depthMap;
        depthMap.convertTo( show, CV_8UC1, scaleFactor );
        imshow( "depth map", show );
        if( waitKey( 30 ) >= 0 )
            break;
    }
    return 0;
}
```

Increasing PandaBoard performance

Here are a few tricks to further increase PandaBoard's performance:

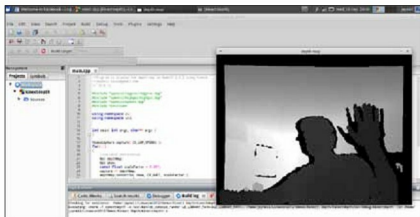


Figure 1: Running example code

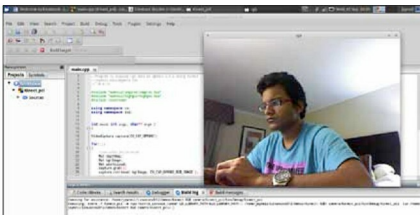


Figure 2: Another example

- Use a lightweight desktop environment such as XFCE, instead of Unity.
- Use an external Solid State Drive (SSD) for faster file access. 

References

- [1] <http://openni.org/docs2/ProgrammerGuide.html>
- [2] http://docs.opencv.org/doc/user_guide/ug_highgui.html

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The author is a FOSS advocate, and loves to explore different open source technologies. His areas of interest include OpenCV, Python, Android, Linux, Human Computer Interaction, BeagleBoard, PandaBoard, Arduino and other open source hardware platforms. In his spare time, he likes to make video tutorials on various open source technologies, which can be found on YouTube. He was a speaker at LinuxCon 2012, Droidcon 2012, and various other international conferences. He is a big-time Arsenal fan. He can be reached at jayneil.dalal@gmail.com.